

THE INTERN DIARIES

Intern Report

Customer Repeat Purchase Analysis

WEEK 4 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

Retail Online II Dataset (CUSTOMER REPEAT PURCHASE ANALYSIS)

1. We spend thousands on ads to get new customers. Is it actually worth it, or should we just focus on keeping the old ones?

Answer: *"Acquisition gets them in the door; Retention makes the profit."*

Acquisition brings customers into the business, while repeat purchasing is associated with higher order value. Our analysis shows that repeat orders have an average order value of \$480.91 compared with \$412.18 for first-time orders, approximately **16.67% higher**. This suggests that converting first-time buyers into repeat customers may increase customer value.

- **The Math:** If it costs \$50 to acquire a customer via ads, and they only buy once (\$320 revenue), the margin is thin. But if we convert them to a repeater, their *next* order is \$480 with **\$0 acquisition cost**.
- **The Verdict:** Ads are worth it *only if* we have a mechanism to convert that first buyer into a second buyer. Without retention, we are renting customers, not owning them. We should shift 20% of the ad budget to "Win-Back" email campaigns for existing users, which typically have a 5–10x higher ROI.

2. Our repeat rate is 34%. Is that good? Should we be worried about the other 66%?

Answer: Repeat Purchase Rate increases substantially as the observation window expands, from **23.61%** within 30 days to **69.50%** within 365 days. In the complete observed customer history, 4,255 of 5,878 customers made at least two purchases, meaning approximately **72.39%** were repeat customers and **27.61% purchased only once**. This indicates a substantial opportunity to encourage one-time customers to return, although the dataset alone does not establish the ROI of a specific retention campaign.

3. Who are our most valuable customers, and how do we make sure we don't lose them?

Answer: The RFM analysis shows strong concentration of customer value among high-value, recently active customers. The Champions segment represents approximately 22% of customers but contributes approximately 68% of total revenue. Champions have high purchase frequency and monetary value combined with strong recency.

This makes them an important retention priority, although the dataset cannot estimate the financial impact of specific loyalty initiatives such as expedited shipping or dedicated account management.

Segment	CustomerCount	CustomerPercentage	AvgRecencyDays	AvgFrequency	AvgMonetaryValue	SegmentRevenue	RevenuePercentage
Champions	1273	21.66	19.00	17.31	9264.65	11793905.74	67.88
Loyal Customers	676	11.50	89.00	7.88	2845.29	1923418.94	11.07
Cannot Lose Them	227	3.86	346.00	8.89	4389.43	996400.17	5.73
Potential Loyalists	633	10.77	24.00	2.91	1319.86	835469.67	4.81
Others	1382	23.51	257.00	1.95	478.43	661189.91	3.81
At Risk	413	7.03	363.00	3.61	1306.26	539484.50	3.10
Needs Attention	695	11.82	444.00	1.36	707.37	491619.46	2.83
Recent Customers	279	4.75	28.00	1.15	319.35	89098.03	0.51

4. I see a segment called 'At Risk'. Why should I care about them? They haven't bought in months.

Answer: The RFM analysis identifies a small group of historically valuable customers who are no longer purchasing frequently. The 'Cannot Lose Them' segment contains approximately 223 customers, representing 3.79% of the customer base and about 5.65% of revenue, with average recency of approximately 341 days. This makes the segment a useful target for re-engagement analysis. However, the dataset does not reveal why these customers became inactive, so specific causes such as competitor switching or poor service cannot be concluded from the available data.

5. Does the type of product they buy first matter? Can we predict who will become a repeater based on their first purchase?

Answer: The current analysis establishes that repeat customers have higher AOV and that repeat behavior varies considerably across customers, but it does not yet establish whether the first product purchased predicts repeat behavior. The Online Retail II dataset contains product descriptions and stock codes but does not provide a ready-made product category suitable for this conclusion. A separate first-purchase product/category analysis would be required before making predictive claims.